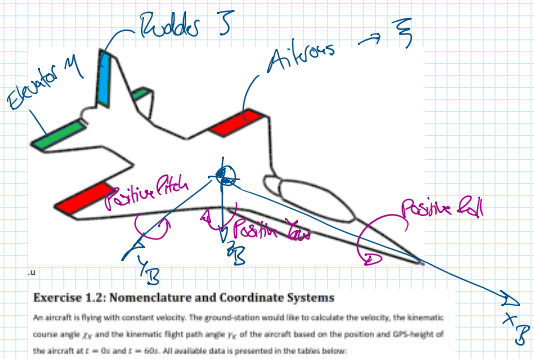
Exercise 1

Mittwoch, 22. Oktober 2025 16:04

Axis	Surface Deflection		Moments	Rate	Attitude	Velocity
x	ξ	xi	L	p	ϕ Phi	u
y	η	eta	M	q	θ Theta	v
z	ζ	zeta	N	r	ψ Psi	w

Exercise 1.1: Aerodynamic Surfaces

Give the symbol and name of the primary aerodynamic control surfaces. Give the symbol and the name of the body fixed axis about which the main moment is produced by each of the surfaces.



Exercise 1.2: Nomenclature and Coordinate Systems

An aircraft is flying with constant velocity. The ground-station would like to calculate the velocity, the kinematic course angle χ_K and the kinematic flight path angle γ_K of the aircraft based on the position and GPS-height of the aircraft at $t = 0s$ and $t = 60s$. All available data is presented in the tables below:

$t_1 = 0$ s:

Geodetic longitude	λ_1	11.669284 deg
Geodetic latitude	μ_1	48.265906 deg
GPS-height	h_1	1500 m

WGS84

$t_2 = 60$ s:

Geodetic longitude	λ_2	11.591421 deg
Geodetic latitude	μ_2	48.22776 deg
GPS-height	h_2	1530 m

You can assume that the earth is *flat* and *non-rotating*. The following steps will guide you through the calculation of the desired quantities.

a.) State the position vectors of both points in the ECEF system

$$x = (N+h)\cdot\cos\mu\cdot\cos\lambda$$
$$y = (N+h)\cdot\cos\mu\cdot\sin\lambda$$
$$z = (N(1-e^2)+h)\cdot\sin\mu$$

$$N = \frac{a}{\sqrt{1-e^2\sin(\mu)^2}}$$

$$e = \sqrt{f(2-f)} = 0{,}0818$$

$$f = \frac{a-b}{a} = 0{,}0034$$

$a \triangleq$ Length of the semi-major axis $= 6378137{,}0$ m

$b \triangleq$ Length of the semi-minor axis $= 6356752{,}3142$ m

$$\left(\vec{r}_1^{\,q}\right)_E = \begin{bmatrix}(x_1)^q\\(y_1)^q\\(z_1)^q\end{bmatrix}_E = \begin{bmatrix}(N_1+h_1)\cos\mu_1\cos\lambda_1\\(N_1+h_1)\cos\mu_1\sin\lambda_1\\(N_1(1-e^2))\sin\mu_1\end{bmatrix} = \begin{bmatrix}4{,}16\cdot10^6\\8{,}6\cdot10^5\\4{,}7\cdot10^6\end{bmatrix}_E \text{ m};$$

$$\left(\vec{r}_2^{\,q}\right)_E = \begin{bmatrix}4{,}17\cdot10^6\\8{,}55\cdot10^5\\4{,}73\cdot10^6\end{bmatrix}_E \text{ m};$$

b.) Calculate the velocity vector of the aircraft by using both position vectors from a). Use a suitable and intuitive coordinate system.

$$\left(\vec{v}^{\,q}\right)_E^E = \frac{1}{\Delta t}\left(\left(\vec{r}_2^{\,q}\right)_E - \left(\vec{r}_1^{\,q}\right)_E\right)$$
$$= \frac{1}{60s}(\dots) = \begin{bmatrix}71.4293\\-83.725\\-46.7151\end{bmatrix}_E^E \text{ m/s};$$

ECEF $\triangleq E$

NED $\triangleq O$

Body-Fixed $\triangleq B$

Inertial $\triangleq I$

E is not intuitive! → Use NED (North, East, Down) $\triangleq O$

$$\left(\vec{v}^{\,q}\right)_O^E = M_{OE}\left(\vec{v}^{\,q}\right)_E^E$$

$$M_{OE} = M_{OA}\cdot M_{AB}\cdot M_{BE}$$
$$M_{EO} = M_{OE}^T$$

$M_{OE} \triangleq$ Frame Transformation Matrix from E to O

$$M_{OE} = \begin{pmatrix}-\sin\mu\cos\lambda & -\sin\mu\sin\lambda & \cos\mu\\-\sin\lambda & \cos\lambda & 0\\-\cos\mu\cos\lambda & -\cos\mu\sin\lambda & -\sin\mu\end{pmatrix}$$

$$\left(\vec{v}^{\,q}\right)_O^E = \begin{bmatrix}-70.6623\\-96.4385\\-0.4328\end{bmatrix}_O \rightarrow \left|\left(v^q\right)_O^E\right| = \sqrt{\left((u^q)_O^E\right)^2+\left((v^q)_O^E\right)^2+\left((w^q)_O^E\right)^2} = 119.556 \text{ m/s};$$

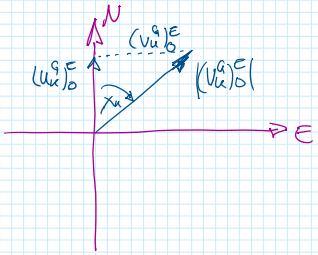
c.) What type of velocity have you calculated in b)?

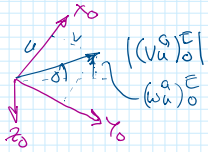
Gravitational Point / C.G.

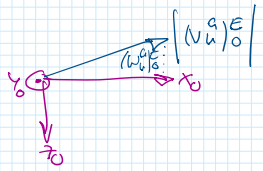
$\left(\vec{V}_K^{\,G}\right)_O^E$ — E ← Frame in which the velocity was computed, ECEF; O ← Notation Frame, NED; K ← Kinematic

d.) Calculate the angles χ_K and γ_K, in point 1 (at t_1).

$$\left(V_K^G\right)_O^E = \begin{bmatrix} u_K^G \\ v_K^G \\ w_K^G \end{bmatrix}_O^E \qquad \left(V_K^G\right)_E^E = \begin{bmatrix} |V_K^G|_O^E| \\ 0 \\ 0 \end{bmatrix} + \chi_K + \gamma_K$$







⊙ goes out of sketch plane
⊗ goes into sketch plane

$$\tan(\chi_K) = \frac{(v_K^G)_O^E}{(u_K^G)_O^E}$$

$$\chi_K = \operatorname{atan}\left(\frac{(v_K^G)_O^E}{(u_K^G)_O^E}\right) = 52.76°$$

$$\sin(\gamma_K) = \frac{-(w_K^G)_O^E}{|(\vec{V}_K^G)_O^E|} \rightarrow \gamma_K = \operatorname{asin}\left(\frac{-(w_K^G)_O^E}{|(\vec{V}_K^G)_O^E|}\right) = 0.207°$$

$$\left(\vec{V}_K^G\right)_O^E = \begin{bmatrix} -71,\dots \\ -87,\dots \\ -46,\dots \end{bmatrix}_O^E \Leftrightarrow \begin{matrix} \text{South} \\ \text{West} \\ \text{Up} \end{matrix}$$

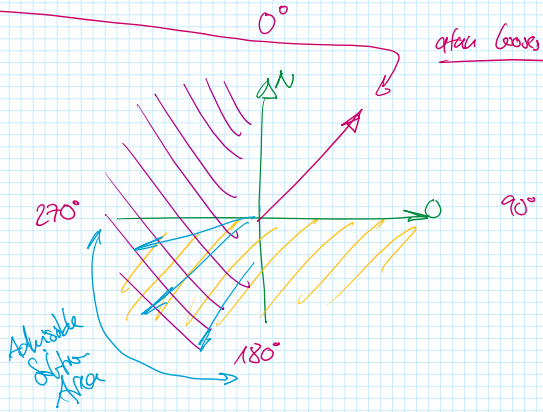
All good!

+180°

$\chi_K = 233.76°$

atan2 → Matlab

atan leaves sign!



e.) Do you find an easy way of checking the results of exercise d) using the results of exercise b)?

Please use the following values in your calculation:

- Length of semi major axis: $a_{earth} = 6378137$ m
- First eccentricity $e_{earth} = 0.0818$

b.): $\left(V_K^G\right)_O^E = \begin{bmatrix} -70.6623 \\ -96.4385 \\ -0.4325 \end{bmatrix}$ m/s → Magnitude: $|(V_K^G)_O^E| = 119.58$ m/s

$\left(V_K^G\right)_E^E = \begin{bmatrix} 71.4293 \\ -87.7215 \\ -46.7158 \end{bmatrix}_E^E$ m/s; → Magnitude: the same!